Literature Review for Master Thesis

1 Avalanche & Snow Modeling

1.1 Micromechanical modeling of snow snow failure (Bobillier et al., 2020)

1.2 Numerical investigation of the mixed-mode failure of snow (Mulak & Gaume, 2019)

Main Question of article: What influence do different loading angles and rates have on failure criteria of snow layers and what effect does sintering have on this?
Why this is relevant for the thesis: Understanding the effect of loading rates, loading angles and failure modes is very important to understand effects seen in DAS data: this influences the type of avalanche, the time between loading and release and whether there is an avalanche at all

Snow material failure resulting in avalanches occurs on the individual bond scale, simultanous shear and compressive loading further increases the complexity (Mulak & Gaume, 2019). Understanding which loading angles and rates lead to which types of failure is of importance in understanding DAS data of avalanche release, because as Mulak and Gaume, 2019 show, these processes are quite complex. Sintering is an important process in understanding material failure, because broken bonds can be reformed, which can even increase the strength of material. Mulak and Gaume, 2019 state that if there is high stress, compression of the material increases, which leads to an increase in strength of the material and the failure is delayed. This has relevance for the thesis because there can be a large amount of time between the loading and the actual material failure. If there is rapid loading on the other hand, sintering is too slow and there is still failure (Mulak & Gaume, 2019). Sintering in the avalanche case would occur if there is a large accumulation of snow and a slow increase in compressive load (Mulak & Gaume, 2019). Could the DAS data look different for rapid loading and slow loading because the stresses are different?

Mulak and Gaume, 2019 implemented a discrete element model where different failure conditItions are investigated at loading angles between 0° and 180°. Failure is defined by an increase in average velocity of the slab (meaning the slab would slide) or an increase in maximum value of the stress (Mulak & Gaume, 2019). Mulak and Gaume, 2019 implemented sintering in their model by the creation of new bonds in the discrete element structure. For simulations with no sintering, at low loading angles, there is a decrease in cohesive bonds, which further increases with higher loading angles (Mulak & Gaume, 2019). This means that at low loading angles, the number of broken bonds required for material failure is much larger, as stated by Mulak and Gaume, 2019. At higher loading angles, the total strength of the weak layer decreases, which means that the shear stress is dependent on loading angle (Mulak & Gaume, 2019). Where sintering was implemented by Mulak and Gaume, 2019, the results are almost the same as in the case without sintering. The number of broken bonds needed for failure is larger however, because new bonds can form, and the strength decreases more if the loading angle increases (Mulak & Gaume, 2019). Mulak and Gaume, 2019 show that sintering is most important in compression-dominant modes at low loading angles. If sintering times are low, the failure of material occurs faster than the formation of new bonds, which means that the results without sintering are the same as the ones without sintering (Mulak & Gaume, 2019). All simulation data show good agreement with lab data. Overall, Mulak and Gaume, 2019 also show that shear strength of a material is more crucial in material failure compared to compressive strength, because shear strength is often lower. This means that the shear load is more likely to cause failure as opposed to the compressive load (Mulak & Gaume, 2019). The models developed by Mulak and Gaume, 2019 are only 2D, which is a limitation because porosity is limited, this limits strain softening and possible collapses of weak layers.

1.3 A physical SNOWPACK model for the Swiss avalanche warning: Part I: numerical model (Bartelt & Lehning, 2002)

Main Question of article: How can the snowpack and its governing equations be modeled accurately but as simple as possible?
Why this is relevant for the thesis: Modeling snowpack evolution and structure is important to

forecast avalanches (where are avalanches likely to form), but also to understand avalanche dynamics (in which layers do avalanches form)

Bartelt and Lehning, 2002 describe that numerical modeling of the snowpack is important for avalanche prediction. The snowpack-model developed solves the partially governing equations using finite-element methods (Bartelt & Lehning, 2002). Understanding snowpack evolution and avalanche formation is also relevant for the thesis because the crack propagation which is modelled in the thesis depends on snow properties. According to Bartelt and Lehning, 2002, most avalanches occur during or after periods of intense snowfall. Snow settling gives an insight on mechanical stability of snow, which is also important to understand crack propagation: in stable layers cracks probably do not propagate as well as in layers which already have some instabilities. The microstructure and the layering of snow also needs to be modeled to detect discontinuities or layer boundaries, which is important to assess stability of layers (Bartelt & Lehning, 2002). Bartelt and Lehning, 2002 also model the influence of solar radiation and warm winds on the energy exchange, the melting of snow and the subsequent transport of meltwater is important to understand the release of wet snow avalanches.

Existing snowpack models such as CROCUS rely on Trusdall's equation for mixture and postulate conservation of ice, water, water vapor and air of the snowpack, these models are simplified (Bartelt & Lehning, 2002). This means that the existing models for example do not describe settling of the snowpack, they assume the snowpack to be dry or the absence of snow metamorphism, which does not depcit the reality of the snowpack accurately (Bartelt & Lehning, 2002). An accurate and physical model of the snowpack is important for the thesis because modelled data should be comparable to field data measured by the DAS cable. This would not be the case if simplifications are made in the simulation of crack propagation in snow, the simulated data would look different from the measured data. The novel SNOWPACK model described by Bartelt and Lehning, 2002 uses three constituents (ice, water and moist air) to describe the snow. These three all have the same temperature, which is the snow bulk temperature measured in meterological observations, assume all meltwater processes to be energy and mass conserving and assume that the weight of the snowpack is carried by a single ice lattice (Bartelt & Lehning, 2002).

Bartelt and Lehning, 2002 use four governing equations to describe the snowpack: the energy transfer equation at the surface (conservation of mass of vapor and water and the bulk temperature), because all of the snow constituents have the same temperature, a momentum equation for settling of the snow because the weight of the snow is carried by a single ice lattice: The conservation of mass of ice is not considered because the numerical solution of the displacement is continuos because of the lagrangian coordinate system. Any change in the snow load results in a change of the stress state of the settlement equation. Snow cannot be assomed to be a plastic material because deformations caused by loading are irreversible, understanding this is important for crack formation and propagation. Bartelt and Lehning, 2002 decsribe a microstructure-based viscosity formulation, plasticity theory is not applicable because of time-dependent yield surfaces and a directionality of plastic flow which is defined in relation to the yield surface. The initial conditions of the snowpack such as snow height are taken from meterological data, the energy exchange at the surface is the most important boundary condition (Bartelt & Lehning, 2002). Neumann boundary conditions are used if the temperature at the surface is close to the melting temperature, if the temperature is well below the melting temperature, Dirichlet boundary conditions are used (Bartelt & Lehning, 2002).

The melting and refreezing processes in the snowpack are essential in modelling snow structure changes, which have an effect on DAS signal propagation asn well as avalanche formation. Bartelt and Lehning, 2002 model these by the numerical heat transfer solution. Because the snow temperature is assumed to be constant 0°C, if the heat transfer solution is greater than 0°C, the excess energy is assumed to be used for melting, if the heat transfer is smaller than 0°C, the energy is assumed to be used for refreezing (Bartelt & Lehning, 2002). Bartelt and Lehning, 2002 used the snow structure measurements of winter 1999, which was a winter with numerous avalanches, to verify the SNOWPACK model developed. Generally, there is good agreement between the measured and modelled data, which means that the model is physical, with some exceptions (Bartelt & Lehning, 2002). If the calculated snow height is different from the measured snow height, the model corrects this by adding elements to the finite element grid in case the calculated height is too low, or removing elements if snow hieght is overestimated (Bartelt & Lehning, 2002). This leads to mistakes if the snowpack settles quickly: if there is quick settlement, snow height decreases quickly because of densification, the model would remove elements at the surface without accounting for densification of the snow at the bottom (Bartelt & Lehning, 2002). Snow density and snow settlement have strong effects on heat conductivity of snow, Bartelt and Lehning, 2002 mention that errors in either of these fields have strong effect on the entire simulation.

The SNOWPACK model developed by Bartelt and Lehning, 2002 includes modeling of the microstructure of snow as well as density discontinuities, which represent layer boundaries, which is important to understand snow structure as well as crack propagation related to avalaches. The multicomponent description developed allows for representation of mass and energy conservation (Bartelt & Lehning, 2002), which can be used to asess snow metamorphism related to temperature and moisture, but also to determine where wet snow avalanches form due to the presence of water. Bartelt and Lehning, 2002 desrcibe the volumetric deformation of fresh snow by large strain description, the thermal boundary conditions which need to be used can be determined by meterological data. These data can also be used to determine if there is melting or refreezing of snow, which again is important for snow structure and stability. The laws for heat conductivity and viscosity related to microstructure show the relations between densification of snow and heat transfer (Bartelt & Lehning, 2002).

1.4 A numerical model to simulate snow-cover stratigraphy for operational avalanche forecasting (Brun et al., 1992)

Main Question of article: How do snow conditions such as temperature gradient and water content influence snow metamporphism and how can this be implemented in the CROCUS-snowpack model?
Why this is relevant for the thesis: Snow metamorphism changes grain properties, which is important for the formation of weak layers which can trigger avalanche release, but also overall stability of the snowpack. Furthermore, layers of different grain properties will depict DAS signals differently.

When a snowpack is subjected to a change, like a change in temperature or pressure conditions, the snow grains change (Brun et al., 1992). Snow metamorphism changes the grain sizes and shapes, which changes mechanical stability of the snow (Brun et al., 1992). This is relevant for avalanche modelling because certain types of snow metamorphism can weaken layers in snow, such layers can act as failure planes and make release of avalanches easier. Snow metamorphism is affected by the water content of the snow, the capillary pressure of the water between the snow grains, the radius of curvature of the grains (shape), as well as the temperature gradient (Brun et al., 1992).

Brun et al., 1992 describes the main limitation of past methamorphism models as assuming the snowpack to have homogenous shapes and sizes, which are often spherical or simplified geometric shapes, this does not represent snow accurately however, especially fresh snow is very heterogenous. Brun et al., 1992 conducted experiments for different snow types and conditions to determine new equations to describe metamorphism. These experiments showed that the rate and type of metamorphism does not depend on crystal type but rather on the temperature gradient (Brun et al., 1992). A gradient below 5°C/m results in rounded crystals, a gradient above 5°C/m results in facteted crystals, which transform to depth hoar, if a gradient of more than 15°C/m is applied (Brun et al., 1992). This is relevant for avalanche modelling, because it is critical to understand how snow grains evolve to asess their stability, which can be used to determine where weak layers will form. These experiments were conducted for dry and wet snow conditions.

Brun et al., 1992 describe a new description of snow, which takes into account the dendricity, sphericity and size of the grains. When snow falls, new layers with different properties are added to the snowpack (Brun et al., 1992). Simulating these different layers is again important to assess the stability as well as interactions of the different layers. Brun et al., 1992 combined the observations made in the laboratory with the CROCUS snowpack model to simulate these different layers based on meterological observations. Using MEPRA (a system for avalanche risk forecasting), the shear stress of the different layers was estimated at a 45° angle. This is important for avalanche forcasting, because the strength of different layers can be assessed and it can be determined under which stress the layers fail.

1.5 A discrete numerical model for granular assemblies (Cundall & Strack, 1979)

Main Question of article: How can the forces and the resulting displacements be modeled in a granular medium?

Why this is relevant for the thesis: Snow consists of grains, forces at the surface such as mechanical loading can cause displacement of the individual grains. If this happens at a very large scale, it is an avalanche.

The distinct element method described by Cundall and Strack, 1979 is a numerical modeling method to model the mechanical behavior of a material composed of disks spheres. This is relevant to the thesis because snow is composed of individual snow grains, which interact with each other at their contacts, all the grains together form the snow pack. It is important to model the behavior and displacement of the grains to model avalanche dynamics because avalanches form through loading or failure in a weak layer, which then propagates through the entire snowpack (Gaume et al., 2017).

Since a snowpack is a medium composed by individual snow grains, which are not entriely compacted (contain air between the grains) and interact with each other at grain contacts, the medium described by Cundall and Strack, 1979 represents snow well. It is a granular medium composed of distinct particles, which interact with each other at their contacts and form a porous medium (Cundall & Strack, 1979). The loading and unloading of such materials is complex and stresses cannot be measured easily because of the interactions of different grains (Cundall & Strack, 1979).

The distinct elememt method can model particles of any shape (Cundall & Strack, 1979), which is important for snow modelling because the snow grains are not always spherical. Cundall and Strack, 1979 model the transient interaction of particles by calculating the contact forces and displacements of the stressed discs (particles) caused by disturbances at the boundaries. In the avalanche case discussed in the thesis, the distrubance is represented by loading of the surface, which can be caused by a skier, which causes failure in a weak layer, or cracking in the snowpack. This results in stress propagation through the snowpack, which results in displacement of the particles, which is the avalanche motion. The resultant foces on the disks are calculated only by the discs which are in direct contact (all the disks that touch one disk cause its resultant force) (Cundall & Strack, 1979). Cundall and Strack, 1979 describe a model only for dry materials, this means this is only applicable for dry snow avalanches which contain no water between the snow grains.

Numerical modelling, such as described by Cundall and Strack, 1979is advantageus over lab tests and experiments because it is less time consuming and any data can be accessed at any stage, which means that parameters such as loading, particle size, and properties such as stiffness and shape can be changed at any stage. In the avalanche case, this means that simulations can be run for different snow properties and loading scenarios, for example a simulation for older, more compacted snow can be run as well as a simulation for fresh snow. The numerical model for spheres is based on methods described by Serrano and Roderiguez-Ortiz (1973, I CANNOT FIND THIS PAPER ONLINE). The forces at the interfaces between the spheres are calculated by incremental displacements of the centers of particles, where shape changes of the individual particles are neglible (Cundall & Strack, 1979). How can the center of the grain be determined if the grain has an irrgular shape? The matrix representing contact stiffness between the grains described by Cundall and Strack, 1979 needs to be recalculated whenever a new contact is made or broken, meaning when there is a crack in the snowpack or when new contacts between snow grains are made. The calculations of forces are based on Newtons 2nd law, which describes the movement of a particle as the result of the forces acting on it. The deformation is neglected because the deformation of individual particles is small compared to the deformation of the medium as a whole (Cundall & Strack, 1979). This means that the deformation of the individual snow grains is small compared to the deformation of the entire snowpack which represents the avalanche movement. What about the fusing / compaction of grains which changes snow properties? The relative displacement of a particle is calculated by the sum of the force increments acting on it, partciles can also overlap, in this case the contact foces and the interaction forces also need to be calculated (Cundall & Strack, 1979). The location of the rigid boundaries (walls) described by Cundall and Strack, 1979 needs to be specified before modelling. This could be a possible limitation for the avalanche case because there is only one rigid "wall" which is the interface of snow and the rock below it.

2 Crack Propagation

2.1 Numerical investigation of crack propagation regimes in snow fracture experiments (Bobillier et al., 2024)

2.2 Transition from sub-Rayleigh anticrack to supershear crack propagation in snow avalanches (trottetTransitionSubRayleighAnticrack2022)

2.3 Dynamic anticrack propagation in snow (gaumeDynamicAnticrackPropagation2018)

2.4 Modeling snow slab avalanches caused by weak-layer failure - Part 2: Coupled mixed-mode criterion for skier-triggered anticracks (Rosendahl & Weißgraeber, 2020)

2.5 Investigating the release and flow of snow avalanches at the slope-scale using a unified model based on the material point method (Gaume et al., 2019)

Main Question of article: How can crack propagation and slab breakoff be modeled using the material point method in 2D and 3D?
Why this is relevant for the thesis: The material point method will also be used in the thesis, understanding how to model slab breakoff with this and which considerations need to be taken is important. Furthermore, it is important to understand how cracks propagate and how this can be modeled to understand and model DAS data of avalanches.

Gaume et al., 2019 state that snow can behave like a solid or like a fluid depending on the type of deformation which is present, modeling this behavior is important to understand which types of loading correspond to which behavior and deformation. The release of avalanches is because if failure initiation in a weak layer, the onset of crack propagation in the weak layer and the resulting detachement of the sliding slab (Gaume et al., 2019). Because the weak layer has porous character, there is volumetric collapse, which leads to the closure of cracks in the weak layer, so called anticrack behavior (Gaume et al., 2019). The new approach to model avalanche dynamics described by Gaume et al., 2019 uses the multi-point method to model deformation. Here, particles in the snowpack are used to track mass, momentum and deformation of the snowpack (Gaume et al., 2019). This means that individual particles instead of fixed mesh points ae used for modeling, the lack of connectivity between the points complicates the calculation of spatial derivatives, so an eularian background is combined with interpolation functions by Gaume et al., 2019 to circumvent this problem. This is also an important consideration in case the MPM method will be used in the thesis. The model described by Gaume et al., 2019 uses a mixed-mode shear and compression yield surface and accounts for softening and hardening of the weak layer. The model accounts for the fact that the internal friction of snow is usually lower than the dynamic friction in the cracks, these are described by different parameters in the model (Gaume et al., 2019).

The simulations run by Gaume et al., 2019 include slope-scale simulations in 2D and 3D with a constant slab depth, which account for spatial variability of the snow depth. The failure in the weak layer is initiated by loading at the bottom of the slope (Gaume et al., 2019). The failure propagates along the weak layer in a mixed-mode anticrack, the propagation speed modeled by Gaume et al., 2019 shows large variations. Changes in slope angle and curvature for example have large influences in the speed: concave leads to high propagation speed, whereas convex parts of the slope have lower speeds (Gaume et al., 2019). The propagation speed of the failure cracks is also influenced by slab tensile fractures: the closer to a slab tensile fracture, the lower the propagation speed (Gaume et al., 2019). The propagation speed of the failure cracks observed by Gaume et al., 2019 also showed strong directionality: propagation in the slope direction (uphill) was observed to be 2x higher than in the downslope direction. Understanding the variations in propagation speed is very important to make sense of DAS observations, for example the time passed between loading and slab breakoff and how this can be explained.

The slab is released wheree the slope angle exceeds the friction angle of the material (Gaume et al., 2019). This is an important consideration when thinking about the observation of slab breakoff in DAS data, because it can be assessed where the breakoff will happen: does it happen far away from the cable when signal could be attenuated or not? Gaume et al., 2019 state that the initial crown fracture of the slab breakoff is followed by a staunchwall and flank shear fractures. Could all of these different fractures be seen in DAS or are they maybe recorded as one event? Broken slab pieces can stick together again, the collective behavior of these particles leads to fluid-like behavior (Gaume et al., 2019). For remote triggering of avalanches, there needs to be the colapse of the weak layer as well as fracture propagation (Gaume et al., 2019). The colapse of the weak layer is neglible on steep slopes according to Gaume et al., 2019, the large angle is sufficient to observe failure by shearing alone. Here, weak layer colapse is secondary to failure Gaume et al., 2019. Could these seperate processes also be observed? Like weak layer colapse would lead to deformation of DAS cable, which could be an indcator even before slab breakoff is observe, but of course only for flatter slopes. Gaume et al., 2019 observed a top to bottom colapse (from top of hill to bottom) in their simulations, which is consistent with large-scale PST experiments. The limitations of the model described by Gaume et al., 2019 is that the strength of the weak layer is not dependent on strain rate. This means that there is no potential for sintering as described by Mulak and Gaume, 2019, the model by Gaume et al., 2019 can therefore only be applied for fast loading, for example for failure induced by a skier. Another limitation of the model is that there is only variation in snow depth, other variations in other parameters are not accounted for by the Gaume et al., 2019 model.

2.6 Snow fracture in relation to slab avalanche release: critical state for the onset of crack propagation (Gaume et al., 2017)

Main Question of article: How can the critical crack length needed for avalanche release be modeled and which parameters influence this?
Why this is relevant for the thesis: The critical crack length and the parameters which influences this are important because it influences what can be seen in DAS data. This is in relation where data is measured: slope angle etc has an influence on how many avalanches are recorded by DAS and subsequently how large the dataset for breakoff observation is

Gaume et al., 2017 develop an equation which describes the relation of critical crack length to snowpack parameters. The release of a dry-snow slab avalanche requires the formation of a weak layer, the initial failure is a macroscopic crack in the weak layer (Gaume et al., 2017). The length of this crack cannot exceed the critical crack length, because else there will be material failure and avalanche release (Gaume et al., 2017). This is because the critical crack length is the length of the crack which is present when the shear stress is equal to the shear strength of the material. Understanding the influences of different snow and terrain parameters on critical crack length is relevant for DAS observations of avalanche breakoff because it can be understood how many avalanches occur and why. Previous modeling of avalanche release to Gaume et al., 2017 made simplifications to crack propagation, which results in inaccurate predictions.

Gaume et al., 2017 estimate critical crack length using a discrete element model as described by Cundall and Strack, 1979, where the cracking is due to sawing in the weak layer, similar to Gaume et al., 2015. The modeled data is then compared to experimental data. The numerical simulations by Gaume et al., 2017 show that critical crack length increases, if elastic modulus of the slab increases or the weak layer thickness increases. The critical crack length on the other hand decreases if the slab density increases, the slab thickness increases or the slope angle increases (Gaume et al., 2017), this means that under these conditions, cracks need to be less long for failure to occur. Gaume et al., 2017 state that the shear stress at the crack tip can be decomposed into a part caused by slab tension and a second part caused by slab bending. The analytical expression for critical crack length taking into account snow and terrain parameters agree with experimental results of saw tests conducted by Gaume et al., 2017. The length of the critical crack is also relevant for the thesis because maybe shorter cracks show up less well in DAS data (less of a displacement). The analytical expression derived describes critical crack length better than the anticrack model, especially in regard to weak layer thickness and slope angle (Gaume et al., 2017). The anticrack model shows that critcial crack length is not dependent on slope angle, this is because the model makes assumptions for simplification which the new expression by Gaume et al., 2017 does not make.

The new expression derived by Gaume et al., 2017 has some limitations, the dependance of critical length on slope angle could also be caused by young's modulus and slab thickness, if these two are decreased, the critical length decreases less with increasing slope angle. Furthermore, Gaume et al., 2017 do not account for the shear stresses caused by slab bending, but suggest that this effect is small because simulation results are in good agreement with observations. The models described here assume a homogenous slab, where in reality the snowpack has multiple layers of different strengths, which can have an effect (Gaume et al., 2017).

2.7 Modeling of crack propagation in weak snowpack layers using the discrete element method (Gaume et al., 2015)

Main Question of article: Which snow parameters influence crack propagation distance and speed and how can these processes be modeled?
Why this is relevant for the thesis: Understanding the influence of different snow parameters on crack propagation distance and speed is important to model and understand slab breakoff DAS data in thesis.

Gaume et al., 2015 state that dry-snow slab avalanches form by the failure of a weak snow layer underlying a cohesive slab. The local damage leading to failure is a crack in the weak layer, which when it reaches critical length leads to tensile fracture in the slab, which leads to avalanche release (Gaume et al., 2015). Understanding how these processes can be modeled and which parameters affect the propagation distance and speed is important for the thesis because slab breakoff is investigated in DAS data. Understanding which snow parameters lead to large crack propagation or speed is important to determine in which DAS datasets slab breakoff is most likely to be detected and at which times in the dataset this would be (does the breakoff take a lot of time or not because speeds and distances are low or high?). Gaume et al., 2015 state that the dynamic phase of crack propagation and the effects of different snow parameters are poorly understood, which limits avalanche forecasting because it is unclear how and under which circumstances cracks propagate. This is directly linked to the thesis, because cracking of the snow layer is investigated in DAS data. The method used by Gaume et al., 2015 to model cracking in snow is the discrete element method described by Cundall and Strack, 1979, with a limitation that the slab is modeled as a continuos material with elastic-brittle behavior.

The loading modeled is due to gravity and the propagation saw test (Gaume et al., 2015), the cohesive part of the snow (so the connections between the grains) are modeled by the linear contact law, the maximum and minimum and minimum tensile and shear stresses are modeled by beam theory (Gaume et al., 2015). The stiffness at the bonded contact (of grains) is calculated (Gaume et al., 2015), this is important to see how the material reacts: what influence does the stiffness of the grain contacts have on crack propagation distance and speed? Gaume et al., 2015 calculate the time-step of the simulations from the snow grain properties, this ensures that the maximum crack propagation velocity is lower than the speed of elastic waves in the snow, this means that the cracks in the snow cannot propagate faster than the propagation of the elastic waves which are caused by the loading. Understanding the relationship between elastic wave velocity and crack propagation speed is also important for the modeling of DAS data of slab breakoff which will be done in the thesis.

Gaume et al., 2015 determined the propagation distance of cracks in the slab by setting the tensile and strength to finite values, to determine propagation speed on the other hand, these values were set to infinity. Understanding the propagation speed and distance of cracks is important because the higher either of these values, the faster the avalanche breaks off. The propagation distance described by Gaume et al., 2015 is defined as the distance between the point where failure initiates, so the point where the initial load is applied and the point where the tensile failure is located. The numerical models showed that the crack propagation speed increases with increasing young's modulus, increasing thickness of slab and increasing density, but the thickness of weak layer has no influence on propagation speed (Gaume et al., 2015). From their results, Gaume et al., 2015 suggest that the lower the young's modulus of a snow layer, meaning the softer a snow layer, the lower the crack propagation velocity, which means that avalanches break off less fast. This suggests that the propagation speed of cracks is mainly influenced by failure conditions such as the tensile strength of the snow layer rather than structural parameters such as weak layer thickness (Gaume et al., 2015), which is relevant for the thesis because factors such as weak layer thickness need not be modeled, which saves computational time. The propagation distance of cracks is influenced by weak layer thickness however, Gaume et al., 2015 suggest that if weak layer thickness increases, there is more slab bending which leads to a higher tensile stress and more likely tensile failure, which means that crack propagation distance decreases, because failure happens before the cracks can propagate very far (Gaume et al., 2015). Furthermore, the simulations run by Gaume et al., 2015 show that the slope angle has little influence on crack propagation for low tensile strength snow layers, but for high tensile strength layers there is a large correlation between slope angle and propagation distance of cracks (Gaume et al., 2015). This can be explained by the main failure mode being tensile failure due to slab bending at high tensile stresses of the snow layer (Gaume et al., 2015).

Gaume et al., 2015 conducted lab experiments, which confirm the results of the numerical simulations in all but one case: the relationship between high young's modulus and large crack propagation is counter-intuitive, because in harder snow, cracks propagate less well compared to soft snow, Gaume et al., 2015 explain this by the limitations of the beam theory to account for dynamic effects. This is an important consideration depending on which method will be used to model crack propagation in the DAS data. Gaume et al., 2015 show that the stress evolution in the snowpack always shows an increase of tensile stress at the top of the slab before crack propagation, which happens close to the tip of the crack in the weak layer. The tensile crack opens up as soon as the tensile stress is equal to the tensile strength of the snow, this crack always starts at the top surface of the slab (Gaume et al., 2015). Understanding the stress evolution of the snowpack is important for DAS data interpretation, because it is possible that the different phases of this path (increase in stress, opening of crack in weak layer, tensile cracking and failure) can be seen in the data. Gaume et al., 2015 mention two important limitations of the simulation results: the models assume a simplified shape of the weak layer, as well as the assumption that the crack propagation speed is lower than the speed of the elastic wave. At thigh propagation speeds of the crack, the tangential displacement of the snowpack becomes more important than the vertical displacement (Gaume et al., 2015). Furthermore, the coulumn length of the modeled propagation saw test, which results in loading of the modeled snow, has large influences on the simulation results depending on the snow properties (Gaume et al., 2015), the thicker and stronger the snow layer, the larger the column must be to get accurate results.

To summarize the snow parameters important for propagation distance and speed, which are important for simulations of slab breakoff in the thesis:

- propagation speed increases if: young's modulus increases, slab depth increases, slab density increases or the slope angle increases
- propagation velocity decreases if weak layer has increasing strength
- propagation distance decreases if young's modulus, slab density or weak layer thickness increases (leads to faster failure)
- propagation distance increases if slab tensile strength, slab depth, weak layer strength or slope angle increases
- For very dense slab layers, tensile fracture and avalanche release is affected by topography (trees, laarge rocks in the avalanche path, or heterogenity of snowcover)

3 Relevant Source Mechanisms

4 Bibliography

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